

Lecture 1 - GOP

25.02.2024

- New I/O streams, exceptions

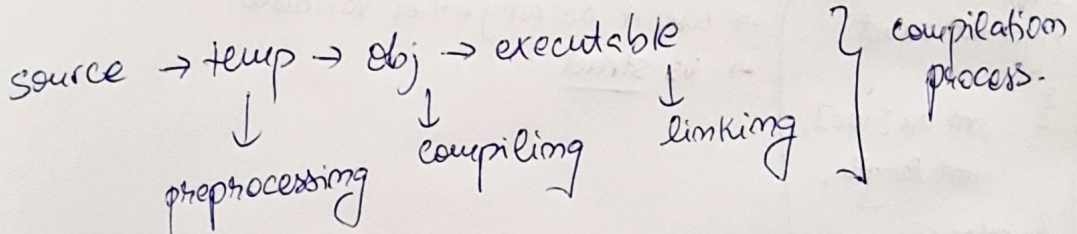
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Unix: Ritchie + Thompson

C++: Stroustrup

Java: Gesling → based on C++.

- "abstract datatype".
- garbage collector (Java)
- Visual Studio 2022 for C/C++.



Compiler translates source code into executable code.

!only select source files when pushing to Git.

char $\rightarrow$ 1 byte

int $\rightarrow$ 4 bytes

double $\rightarrow$ 8 bytes

variable = named memory location

pointer = a variable whose name is a memory location (address)

int *p → pointer declaration. → p contains inside the address of
char *str → → another char another integer.

$\&p \rightarrow$ address of p

int a = 7

&a → address of a → ampersand

*a → take the value at the address of the pointer. → dereference

! See behaviour of pointers in C vs C++.

• Array = continuous block of memory which contains values...

mat[m][m] → multidimensional array.

! addresses of the elements in the array → consecutive nr.

name of array = a pointer to the first elem. in the array.

(address of b[0] = address of b)

b[3] → value found at address b+3 is... * (b+3).

||

3[b]

typedef → define new type (introduce a new type).

typedef struct

```
{  
    int arr[100];  
    int length;
```

```
} vector;
```

→ use it as any other variable

→ vs struct

typedef integer

no strings in C → use characters of characters.

string.h

scanf-s → not allowed to put too big stuff → not recommended because they aren't in the standard library.